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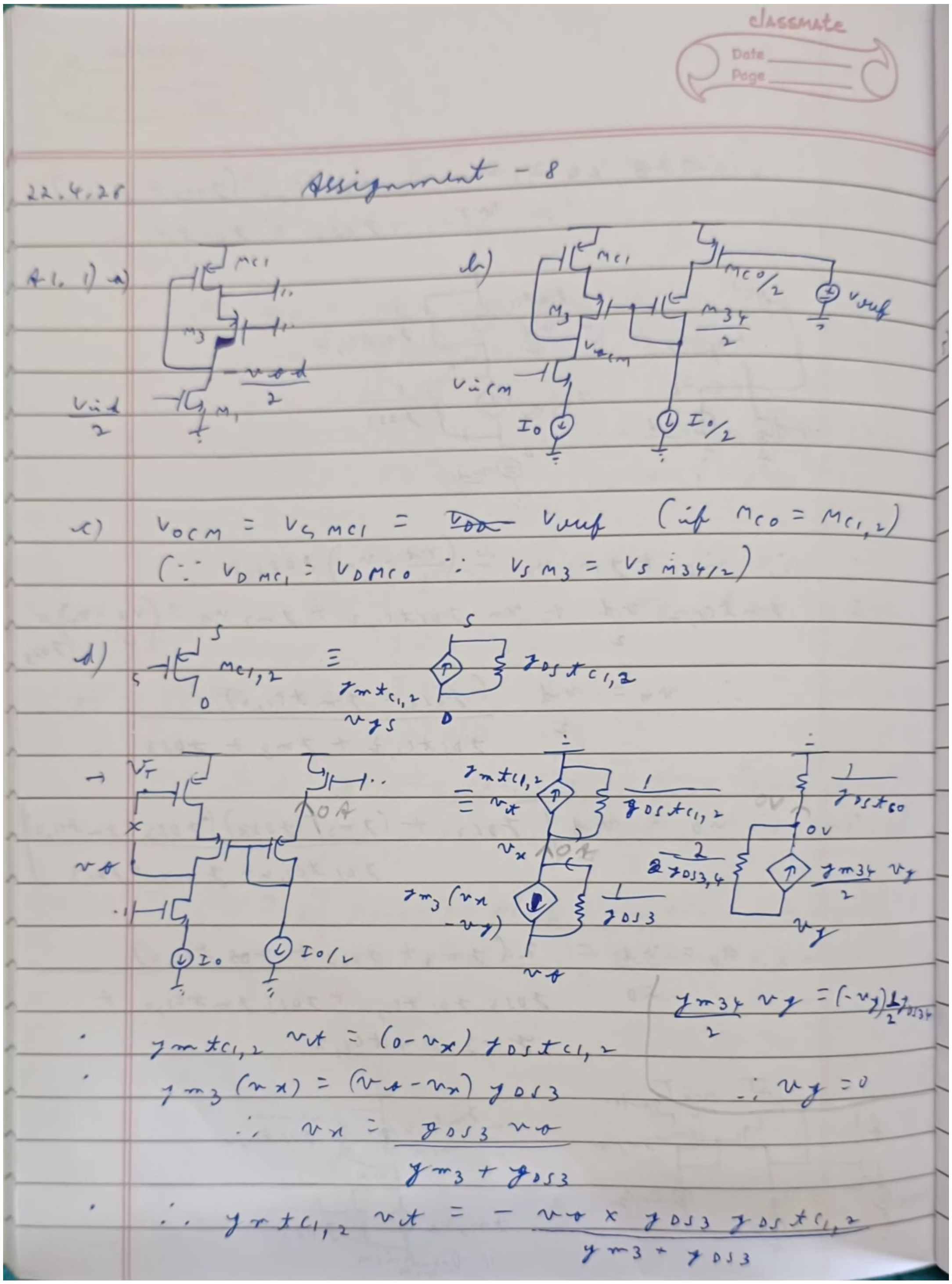
Roll number	EP22B020

8.1:



22.4.26

Assignment - 8

8.1. 1) a)

b)

c) $V_{OCM} = V_{S\,MC1} = \cancel{V_{DD}}\ V_{ref}$ (if $M_{C0} = M_{C1,2}$)

($\because V_{D\,MC1} = V_{D\,MC0}$ $\therefore V_{S\,M3} = V_{S\,M34/2}$)

d) $M_{C1,2} \equiv g_{mtc1,2}\, v_{gs}$, $r_{dstc1,2}$

$\frac{g_{m34}}{2} v_y = (-v_y)\frac{1}{2} g_{ds34}$

$\therefore v_y = 0$

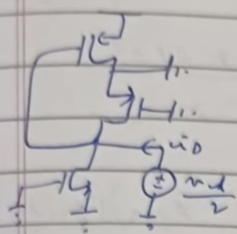
- $g_{mtc1,2}\, v_t = (0 - v_x)\, g_{dstc1,2}$
- $g_{m3}(v_x) = (v_o - v_x)\, g_{ds3}$

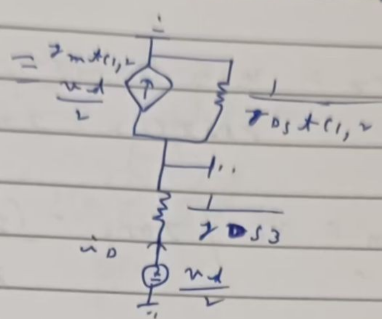
$$\therefore v_x = \frac{g_{ds3}\, v_o}{g_{m3} + g_{ds3}}$$

- $$\therefore g_{mtc1,2}\, v_t = -\frac{v_o \times g_{ds3}\, g_{dstc1,2}}{g_{m3} + g_{ds3}}$$

$$\therefore CM \neq 0,\ A_c = \frac{v_o}{v_+} = -\frac{g_{m,tc1,2}\,(g_{m3} + g_{ds3})}{g_{ds3} \times g_{ds,tc1,2}}$$

e)

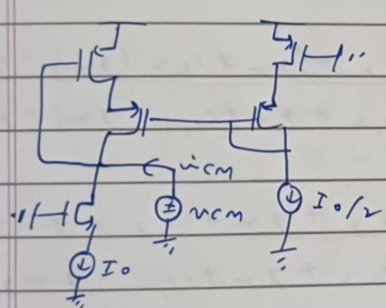


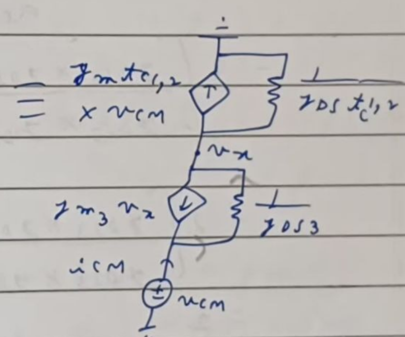


$$\therefore \left(\frac{v_d}{2} - 0\right) g_{ds3} = i_D$$

$$\therefore a_D = \frac{v_D}{i_D} = \frac{2}{g_{ds3}}$$

f)





- $i_{cm} + g_{m3} v_x = (v_{cm} - v_x)\, g_{ds3}$
- $(v_{cm} - v_x)\, g_{ds3} = g_{m,tc1,2} \times v_{cm} + g_{m3} v_x + v_x \times g_{ds,tc1,2}$

$$\therefore v_x = \frac{v_{cm}\,(g_{ds3} - g_{m,tc1,2})}{g_{ds3} + g_{m3} + g_{ds,tc1,2}}$$

$$\therefore\ i_{cm} = v_{cm}\left[g_{ds3} + \frac{(g_{m\,tc1,2} - g_{ds3})(g_{m3} + g_{ds3})}{g_{ds3} + g_{m3} + g_{ds\,tc1,2}}\right]$$

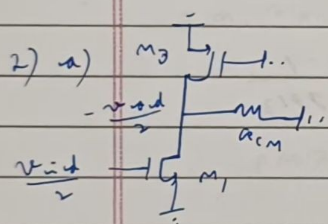
$$\therefore\ R_{cm} = \frac{v_{cm}}{i_{cm}} = \frac{g_{ds3} + g_{m3} + g_{ds\,tc1,2}}{g_{ds3} \times g_{ds\,tc1,2} + g_{m\,tc1,2}(g_{m3} + g_{ds3})}$$

g) CMRR, CMRR $= \dfrac{R_D}{R_{CM}}$

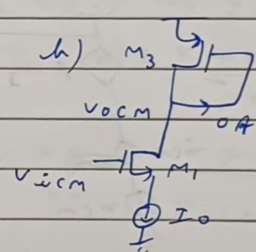
$$= \frac{2}{g_{ds3}}\left[\frac{g_{ds3}\, g_{ds\,tc1,2} + g_{m\,tc1,2}(g_{m3} + g_{ds3})}{g_{ds\,tc1,2} + g_{m3} + g_{ds3}}\right]$$

$$\simeq \frac{2}{g_{ds3}}\left[\frac{g_{ds3}\, g_{ds\,tc1,2} + g_{m\,tc1,2}\, g_{m3}}{g_{ds\,tc1,2} + g_{m3}}\right]$$

2) a)

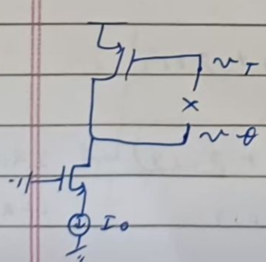


b)



c) $\therefore\ V_{OCM} = V_{GS\,m3,4} = V_{DD} - V_{GS\,3,4}\,|\,I_0$

d)



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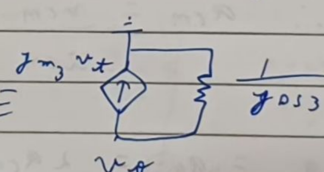
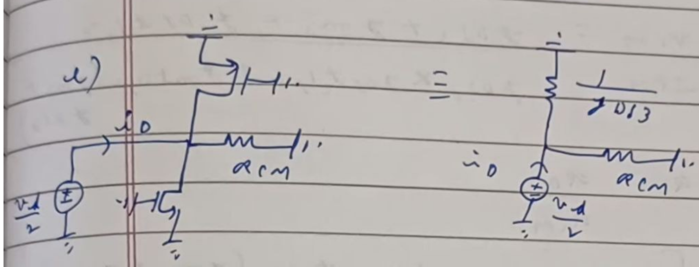


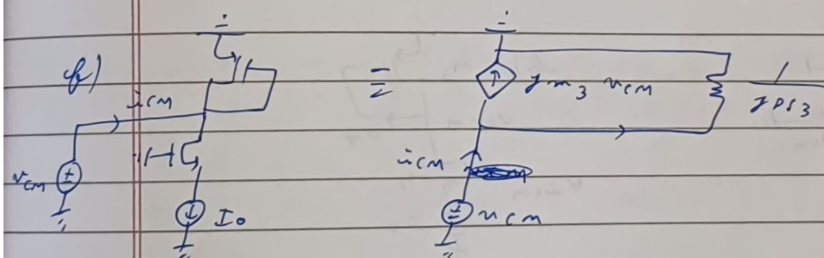
$\therefore g_{m3} v_t = (0 - v_t) g_{DS3}$

$\therefore$ CMFB, $A_L = \dfrac{v_o}{v_t} = \dfrac{-g_{m3}}{g_{DS3}}$

e) 

$$\therefore \frac{v_d}{2} = i_D \times \left(R_{CM} // \frac{1}{g_{DS3}} \right)$$

$$\therefore R_D = \frac{v_d}{i_D} = 2 \frac{R_{CM}}{1 + R_{CM} g_{DS3}}$$

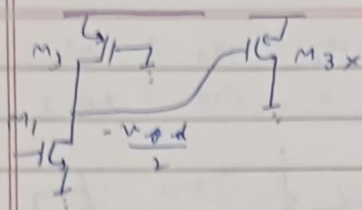
f) 

$$\therefore i_{CM} = v_{CM} g_{DS3} + g_{m3} v_{CM}$$

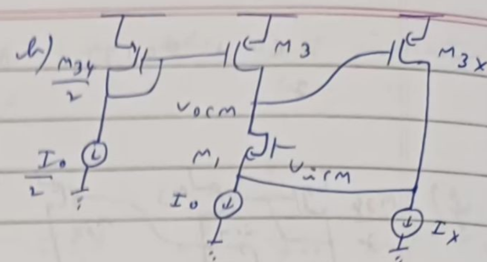
$$\therefore R_{CM} = \frac{v_{CM}}{i_{CM}} = \frac{1}{g_{m3} + g_{DS3}}$$

g) $\therefore \dfrac{\text{CMFB}}{\text{CMRR}} = \dfrac{R_D}{R_{CM}} = \dfrac{2 R_{CM} (g_{m3} + g_{DS3})}{1 + R_{CM} g_{DS3}} \simeq \dfrac{2 R_{CM} g_{m3}}{1 + R_{CM} g_{DS3}}$

3) a)

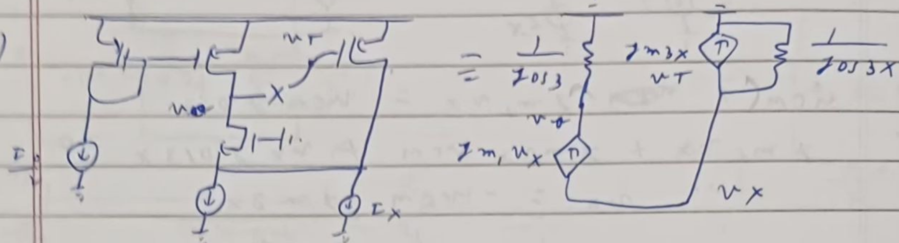


b)



c) $V_{OCM} = V_{G\,M3x,4x} = V_{DD} - V_{SG\,3x,4x}|_{I_x}$

d)



$g_{m1} v_x = v_o\, g_{ds3} \Rightarrow v_x = v_o \times \frac{g_{ds3}}{g_{m1}}$

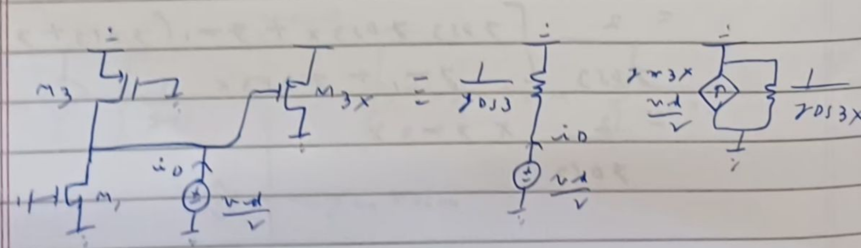
$g_{m1} v_x + g_{m3x} v_T + v_x\, g_{ds3x} = 0$

$\therefore -g_{m3x} v_T = v_o \times \frac{g_{ds3}}{g_{m1}} (g_{m1} + g_{ds3x})$

$$\therefore \text{CMFB, LG} = \frac{v_o}{v_T} = \frac{-g_{m3x}\, g_{m1}}{g_{ds3}(g_{m1} + g_{ds3x})}$$

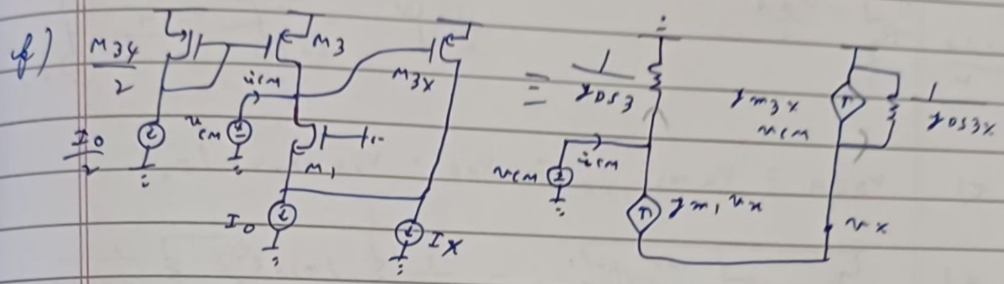
$$\simeq -\frac{g_{m3x}}{g_{ds3}}$$

e)



$$\frac{v_d}{2} \times g_{DS3} = i_D \Rightarrow a_D = \frac{v_D}{i_D} = \frac{2}{g_{DS3}}$$

f)



- $i_{cm} + g_{m1} v_x = v_{cm}\, g_{DS3}$
- $g_{m1} v_x + g_{m3x} v_{cm} + v_x\, g_{DS3x} = 0$

$$\therefore v_x = - v_{cm} \frac{g_{m3x}}{g_{m1} + g_{DS3x}}$$

- $$\therefore i_{cm} = v_{cm}\left[g_{DS3} + \frac{g_{m3x}\, g_{m1}}{g_{m1} + g_{DS3x}}\right]$$

$$\therefore a_{cm} = \frac{v_{cm}}{i_{cm}} = \frac{g_{m1} + g_{DS3x}}{g_{DS3}\, g_{DS3x} + g_{m1}(g_{DS3} + g_{m3x})}$$

g)
$$\therefore \frac{CMRR}{CMRR} = \frac{a_D}{a_{CM}}$$

$$= \frac{2}{g_{DS3}}\left[\frac{g_{DS3}\, g_{DS3x} + g_{m1}(g_{DS3} + g_{m3x})}{g_{m1} + g_{DS3x}}\right]$$

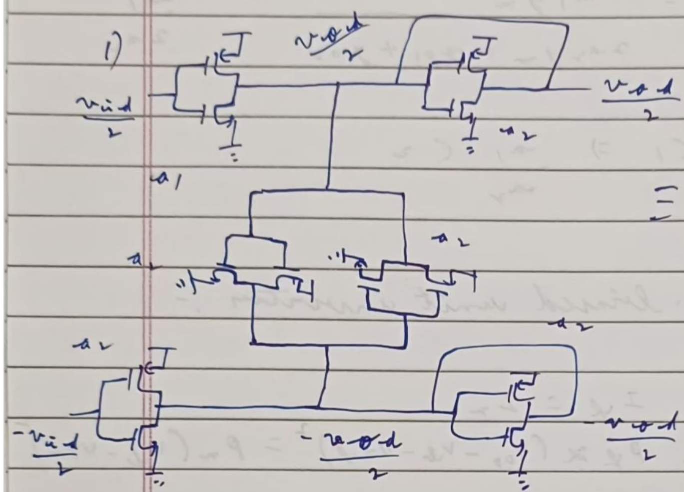
$$\simeq \frac{2}{g_{DS3}} \times g_{m3x}$$

8.2:

A2. a) $g_m = g_{mn} + g_{mp}$ of the unit inverter

$g_o = g_{dsn} + g_{dsp}$ " " " "

1)



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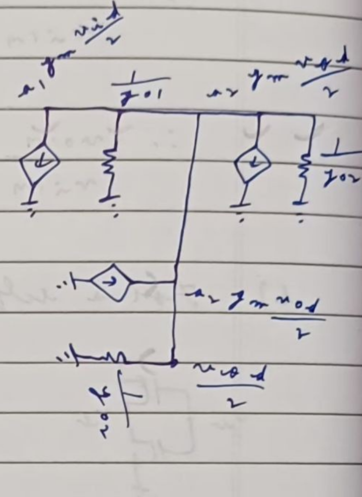
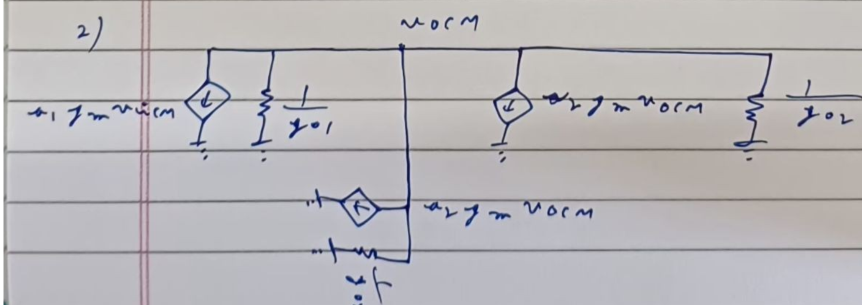


$$\therefore \cancel{a_2 g_m \frac{v_{od}}{2}} = \cancel{a_2 g_m \frac{v_{od}}{2}} + a_1 g_m \frac{v_{id}}{2} + \frac{v_{od}}{2}\left(g_{o1} + 2g_{o2}\right)$$

$$\therefore \frac{v_{od}}{v_{id}} = \frac{-a_1 g_m}{g_{o1} + 2g_{o2}}$$

2)

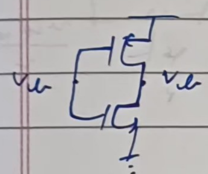


$$\therefore a_1 g_m v_{icm} + 2a_2 g_m v_{ocm} = v_{ocm}(g_{o1} + 2g_{o2})$$

$$\therefore \frac{v_{ocm}}{v_{icm}} = \frac{-a_1 g_m}{2a_2 g_m + g_{o1} + g_{o2}} \simeq -\frac{a_1}{2a_2}$$

$$\therefore \frac{v_{ocm}}{v_{icm}} < 1 \Rightarrow \frac{a_1}{a_2} < 2$$

b) For a self-biased unit inverter :-



$$I_p = I_n$$

$$\therefore \beta_p x (V_{DD} - V_b - V_{Tp})^2 = \beta_n (V_b - V_{Tn})^2$$

$$\therefore I_0 = \frac{1}{2}\beta_n (V_b - V_{Tn})^2 = I_n = I_p$$

$$\therefore I_{net} = (2a_1 + 4a_2) I_0 \qquad (\because \beta \propto W)$$